

Master Thesis Supervisor Meeting – Briefing Agenda

Mode-I Phase-Field Convergence, UEL Energy Formulation & Adaptive Localization

Date: Thursday, 08 October 2026, 10:00 · **Duration:** 45 Minutes

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Governing Supervisor Directive & Active Phase

“We need to have understood everything related to the first model before we increase complexity.”

Meeting Agenda & Discussion Flow (45 Minutes)

1. Opening Decision: Mode-I Adaptive Localization & Step Selection (00–08 min)

- **Executive Decision Sheet:** Root cause of downstream coarsening isolated to Step-1 vs Step-2 **RemeshingRule** evaluation.
- **Quantitative Evidence:** Step 2 increases corridor elements by +181.8% (3,351 → 9,442), cuts wake waste by −75.9%, and maintains $h_A \leq 1.5 \mu\text{m}$ across the ligament (62,057 finite elements).
- **Epistemological Boundary:** Project root cause verified; Pandey–Kumar step argument remains unknown.
- **Supervisor Ruling Requested:** Option A (adopt Step 2 as baseline) vs Option B (retain Step 1 as reproduction, Step 2 as project correction). Qualification Job 1409585.mm_{master02} audited: 87.9% load drop across 94.0% ligament; breakthrough cutback classified as **ROOT_CAUSE_CANDIDATE_NOT_YET_RECONCILED**.

2. Priority 1: UEL Energy Formulation & Output Audit (08–18 min)

- Non-invasive energy exposure in **f42_mixed_uel.for**: exact bit-for-bit mechanical parity ($|\Delta F| = 0.0 \text{ kN}$).
- Single-IP extraction rule: eliminated 4× overcounting artifact on companion visualizer CPE4 elements.
- Disproven common discrete potential ($\partial^2 \Pi / \partial \mathbf{u} \partial d \neq \partial^2 \Pi / \partial d \partial \mathbf{u}$).
- Observability boundary: Newton paths unpersisted; status **GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED**.

3. Priority 2: Multifaceted Mode-I Convergence Findings (18–28 min)

- Initial stiffness K_0 : completely stable across spatial meshes ($< 0.09\%$) and time steps ($< 0.001\%$) (**CONVERGED / STABLE**).
- Peak force $F_{\max}$: strictly stable temporally ($< 0.07\%$), mesh-sensitive spatially (4.26% drop across $h/l_0 \in [0.13, 0.40]$).
- Spatial energetic convergence: post-peak E_{frac} varies by 1.56% across S_1 – S_4 (2.339 → 2.375 mJ, **STABLE_OVER_TESTED_RANGE**).
- Adaptive remeshing series (A_1 – A_4): pre-peak stiffness matched; post-peak elevated plateau on coarse background meshes.

4. Crack-Path Symmetry, Localization & Length-Scale Study (28–36 min)

- Crack centroid trajectory: damage-weighted offset $|y_c - 0.5 \text{ mm}| \leq 3.10 \mu\text{m}$ (element-bounded pure Mode-I symmetry).
- Regularized localization band: full transverse width $w_{d \geq 0.5} \approx 20.0$ – $30.0 \mu\text{m}$ (mesh-sensitive and resolution-limited).
- Completed 3-point length-scale sensitivity study on S_3 mesh ($l_0 \in \{7.5, 11.25, 15.0\} \mu\text{m}$, Jobs 1406017, 1406895, 1406896).

5. Thesis Integration & Gate 6C Promotion Decisions (36–45 min)

- Approval of completed Mode-I thesis report chapters (Chapters 1–3, 7–8).
- Ruling on Gate 6B closure and acceptance of **GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED**.
- Authorization to proceed with Gate 6C (Mode-I State-Transfer Energy Conservation).
- Reaffirmation of holds on Mode-II, multi-crack models, and Gate 7 visualization.